

Instagram User Analytics

Description:

This project is based on analyzing a social media platform which is Instagram and understanding platform activities along with user behaviour and engagement.

SQL queries are used for filtering data and extracting insights which helps in marketing, improving user experience and decision making for investors.

Approach:

Firstly, I created a database in the MySQL workbench using the given SQL dataset file where all the user information is present. Then I start to explore the tables such as users, likes, photos, etc., while understanding how the given data is structured.

Then I start cleaning the data by concentrating on each business issue and start writing the SQL queries to filter the raw data and solve them using joins, aggregations, etc.

While solving each problem I validated and documented the outputs with proper insights regarding each and every issue which can help the business grow.

Tech-Stack Used:

MySQL Workbench 8.0.45 (Windows) - Used for creating database and writing SQL queries for filtering and analyzing the data to find solutions to the issues.

Microsoft Word/PDF - Used for documentation and making report of the outputs and insights for the future developments.

Insights:

Marketing Analysis

Loyal User Reward –

```
113 • select * from users
114 order by created_at asc
115 limit 5;
```

id	username	created_at
80	Darby_Herzog	2016-05-06 00:14:21
67	Emilio_Bernier52	2016-05-06 13:04:30
63	Elenor88	2016-05-08 01:30:41
95	Nicole71	2016-05-09 17:30:22
38	Jordyn.Jacobson2	2016-05-14 07:56:26
NULL	NULL	NULL

SQL Query

Output

According to the outputs, there are the top 5 earliest users who have been using the platform for the longest time.

They could be awarded as the loyal users as they have more trust on the platform. For more engagement, loyalty perks could be provided such as exclusive badges.

It is important to retain oldest users as they bring value and influence platform culture.

Inactive Users Engagement -

```
119 • select users.id, username from users
120 left join photos
121 on users.id = photos.user_id where photos.id is null;
122
```

SQL Query



The screenshot shows a 'Result Grid' with columns 'id' and 'username'. It lists 20 users who have not posted any photos. The users are: Anya_Hackett, Kasandra_Homenick, Jacyln81, Rocio32, Maxwell_Holmstrom, Tierra_Trambow, Pearl7, Olle_Ledner37, McKenna17, David_Corral47, Morgan_Kassulke, Linnea59, Duane60, Julien_Schmidt, Mike_Auer29, Franco_Kuebler64, Nia_Haag, Hulda_Macejkovic, Leslie7, Janelle_Hillman81, Darby_Herzog, Esther_Zulauf61, Bartholome_Bernhard, Janyra_Veal, Emeraldia_Mraz57, and Bethany20.

id	username
5	Anya_Hackett
7	Kasandra_Homenick
14	Jacyln81
21	Rocio32
24	Maxwell_Holmstrom
25	Tierra_Trambow
34	Pearl7
36	Olle_Ledner37
41	McKenna17
45	David_Corral47
49	Morgan_Kassulke
53	Linnea59
54	Duane60
57	Julien_Schmidt
66	Mike_Auer29
68	Franco_Kuebler64
71	Nia_Haag
74	Hulda_Macejkovic
75	Leslie7
76	Janelle_Hillman81
80	Darby_Herzog
81	Esther_Zulauf61
83	Bartholome_Bernhard
89	Janyra_Veal
90	Emeraldia_Mraz57
91	Bethany20

Output

According to the outputs, these are the users who never posted any photos on the platform which indicates they are registered but not engaged.

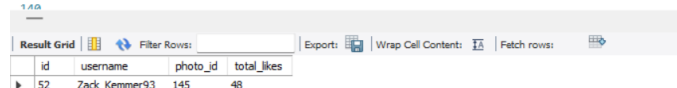
This could be possible because of no exciting motivation which are missed opportunities.

For more engagement, promotional mails could be sent featuring new programs which involves first post benefits or limited features for users who would be posting for the first time.

Contest Winner Declaration -

```
133 • select users.id, username, photos.id as photo_id, COUNT(likes.user_id) as total_likes
134 from photos
135 join likes on photos.id = likes.photo_id
136 join users on photos.user_id = users.id
137 group by photos.id
138 order by total_likes desc
139 limit 1;
```

SQL Query



The screenshot shows a 'Result Grid' with columns 'id', 'username', 'photo_id', and 'total_likes'. It displays the top result for the contest winner.

id	username	photo_id	total_likes
52	Zack_Kemmer93	145	48

Output

According to the output, this is the present user with most number of likes on a single photo, hence the contest winner.

Based on this we can analyze the type of photo, users who are liking it reflecting their interests and promoting more such contents as they provide ways for future engagement strategies.

Hashtag Research -

```
143 • select tag_name, count(photo_tags.photo_id) as total_usage
144 from tags
145 join photo_tags on tags.id = photo_tags.tag_id
146 group by tags.id
147 order by total_usage desc
148 limit 5;
```

SQL Query

tag_name	total_usage
smile	59
beach	42
party	39
fun	38
concert	24

Output

These are the top five most commonly used hashtags on the platform which indicates current trends among users and also support brands to optimize more reach and visibility.

For more engagement, creating separate section for trending hashtags are recommended which could guide brand campaign and amplify their reach.

Ad Campaign Launch –

```
152 • SELECT DAYNAME(created_at) AS day, COUNT(*) AS total_users
153 FROM users
154 GROUP BY day
155 ORDER BY total_users DESC;
```

SQL Query

day	total_users
Thursday	16
Sunday	16
Friday	15
Tuesday	14
Monday	14
Wednesday	13
Saturday	12

Output

According to the output, the day with most number of user registration will be the best day to launch Ad campaign.

As on specific day there are most number of user engagement and signups, so launching Ads a day prior to that could benefit in maximizing awareness and engagement.

Investor Metrics

User Engagement -

```
164 • select avg(photo_count) as avg_posts_per_active_users
165 from (
166 select user_id, count(*) as photo_count
167 from photos
168 group by user_id
169 ) as user_posts;
```

SQL Query

avg_posts_per_active_users
3.4730

Output

According to this output, it shows the average number of posts per user who have posted and are active.

```
158
159 • select
160     (select count(*) from photos) / (select count(*) from users)
161     as Avg_posts_per_all_users;
162
```

SQL Query

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: [IA](#)

Avg_posts_per_all_users
2.5700

2.5700

Output

This output shows the average amount of posts across all users on the platform.

Here the outputs represent the health, user activity and stability of the platform which shows that all the users on the platform are not posting and many users are inactive. Few core users are highly active where as others are not engaging.

For more growth and engagement, strategies such as remainder mails, exciting offer campaigns for loyalty and first post benefits should be implemented focusing on the inactive users.

Bots & Fake Accounts -

<pre>169 • select id, username 170 from users 171 join likes on id = user_id 172 group by id, username 173 having count(photo_id) = (select count(*) from photos); 174</pre>	SQL Query																												
Result Grid Filter Rows: Export: Wrap Cell Content: <table><tr><th>id</th><th>username</th></tr><tr><td>5</td><td>Aniya_Hackett</td></tr><tr><td>14</td><td>Jadyn81</td></tr><tr><td>21</td><td>Rocio33</td></tr><tr><td>24</td><td>Maxwell.Halvorson</td></tr><tr><td>36</td><td>Ollie_Ledner37</td></tr><tr><td>41</td><td>Mckenna17</td></tr><tr><td>54</td><td>Duane60</td></tr><tr><td>57</td><td>Julien_Schmidt</td></tr><tr><td>66</td><td>Mike.Auer39</td></tr><tr><td>71</td><td>Nia_Haag</td></tr><tr><td>75</td><td>Leslie67</td></tr><tr><td>76</td><td>Janelle.Nikolaus81</td></tr><tr><td>91</td><td>Bethany20</td></tr></table>	id	username	5	Aniya_Hackett	14	Jadyn81	21	Rocio33	24	Maxwell.Halvorson	36	Ollie_Ledner37	41	Mckenna17	54	Duane60	57	Julien_Schmidt	66	Mike.Auer39	71	Nia_Haag	75	Leslie67	76	Janelle.Nikolaus81	91	Bethany20	Output
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According to the output, these are the users who have liked every single post on the platform which typically is not possible for a normal user which indicated to the suspicion of bots or fake accounts.

If avoided, this could lead to misleading data and the analytics. For the maintenance of data integrity and trust it is essential to identify these accounts where AI bot detection could be used also with reporting and marking accounts which seem suspicious.

Result:

From this project I provided meaningful insights which helped the teams optimize campaigns and identify engagement of trends leading to understand user behaviour and improving strategies for better business growth.

Enhancing investor trust and confidence by highlighting and providing insights on platform health along with detection of potential risks to ensure better reliability and authenticity.

This project helped me get the knowledge of how SQL could be used for transforming raw data into insights for business decision making while enhancing practical understanding on real world application.

Drive Link:

https://drive.google.com/drive/folders/1kCq8vL_yteJqaOEsSCC3IcZ9RhK-rjqB?usp=sharing